

# EE2000: Signal Processing, Fall 2026

Indian Institute of Technology Hyderabad  
Course Project Flier

## Guidelines and Instructions:

- Please form *four-member* teams.
- Scores on all the project components will be awarded team-wise, i.e., the team's score will apply to all the team members.
- Team formation deadline: *05.09.2026*
- Preliminary report - *27.09.2026*: This report should include a clearly identified problem statement, and a summary of the related literature reviewed. The report will follow the standard IEEE CVF Conference template. This template can be downloaded from here. This report carries 5% of the total grade.
- Mid-term report - *25.10.2026*: This report should include a summary of any further literature review, results replicated from the literature, and preliminary results from your work. Any code developed/experimented with must be submitted. This report carries 5% of the total grade.
- Final report - *13.11.2026*: This report should build on the previous reports and present your algorithm, results and conclusions. Your code must be submitted along with this report. This report + results + code carries 5% of the total grade. The work done by each team member must be clearly delineated and the code written by each team member must be appropriately commented.
- Project presentation and viva - *TBD*: The team must present their work on the *Boolean Board*, and answer questions. The exact schedule for this will be shared closer to the presentation date. The presentation and viva will account for 5% of the grade.
- AI usage policy: You are free to use AI tools such as ChatGPT, Claude, etc. to help you with your project. However, you must clearly indicate the parts of the work that were generated by AI tools and provide appropriate citations. Further, you must be able to explain and defend the AI-generated content during your project presentation and viva.

## A non-exhaustive list of topics:

- Signal processing for:
  - Biomedical applications, e.g., ECG, EEG, EMG, etc.
  - Audio and speech processing, e.g., speech enhancement, music source separation, etc.
  - Image and video processing, e.g., image denoising, super-resolution, etc.
  - Communication systems, e.g., channel equalization, MIMO detection, etc.
- You are also welcome to choose from current trends in machine learning for signal processing.